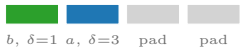


Timeline

 $t-3$ $t-2$ $t-1$ t (query)

Dyad Stream Encoder

stream of (s, o_1) , most recent firstrelation embedding $\mathbf{E}_\rho[\rho_i]$ time code $\phi(\delta_i)$
sin, cos of $\log(1+\delta_i)$ read-out $\mathbf{q}_r$
 $\mathbf{E}_q[r] + \mathbf{W}_s \mathbf{E}_t[s]$

multi-head self-attention

feed-forward block 1

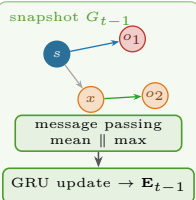
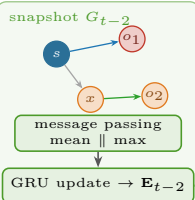
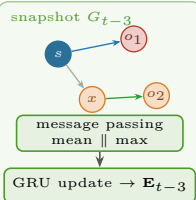
multi-head self-attention

feed-forward block 2

stream state $\mathbf{z}_{so}$

one sequence per (query, candidate); the order and the types of the events enter here

Stream Encoder

Dyad streams of the support $S(q)$ typed, timed tokens (ρ_i, δ_i) , one row per candidate, built from the events before t 8 statistics $\mathbf{a}_{so}$
counts, recency, mean gapcontext $\mathbf{c}_o$
the candidate's own edgesCandidate states $\mathbf{H}_q$ rows o_1, o_2, x $[\mathbf{z}_{so}; \text{LN}(\mathbf{a}_{so}); \mathbf{W}_e \mathbf{E}_t[o]; \mathbf{c}_o; \mathbf{q}_s; \mathbf{q}_r]$ attention across the candidates of q 

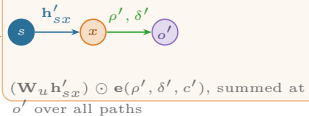
3x3 attention: each candidate sees the others' streams

refined states $\mathbf{H}'_q$ evolved states $\mathbf{E}_t$

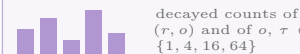
Intensities and ranking

$$\log \lambda_{\text{dyad}}(o) = \mathbf{w}^\top \mathbf{h}'_{so} + b_{\text{dyad}}, \text{ on } S(q)$$

path propagation

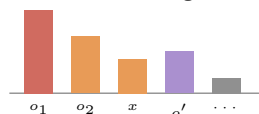
 $\log \lambda_{\text{path}}(o')$, two hops from s

popularity field

 $\log \lambda_{\text{pop}}(o)$, all N entities $\log \lambda_{\text{struct}}(o)$, all N ConvTransE($\mathbf{E}_t[s], \mathbf{R}[r]$) $\cdot \mathbf{E}_t$

$$\sum \leftarrow \log \text{addexp}$$

score ranking



Superposition and decoding

Evidence construction and dyadic encoding